

Mapping Earthquake Origin and Ground-Shaking Intensity in Pakistan, 1900-2025

Data Sources, Software Stack, Processing Pipeline, and Interpretive Analysis

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Abstract

This report documents the construction and interpretation of two companion cartographic products for Pakistan: an oblique three-dimensional relief map of earthquake epicentres ($M \geq 2.5$, 1902–2025) and a companion map of maximum ground-shaking intensity (2000–2025). The distinction between the two is the central methodological point. An epicentre map records where ruptures originated; it is not a record of where shaking was experienced. Because macroseismic intensity attenuates with distance, densely populated lowlands can undergo damaging shaking while registering almost no local epicentres. The analysis quantifies this gap: the Punjab plain contains 61 catalogued epicentres against 840 in the Sulaiman–Kirthar fold belt at identical latitudes, yet 98.4 per cent of Pakistan's land area has experienced at least Modified Mercalli Intensity IV in the past 25 years, and 28.9 per cent has experienced MMI VI or greater. The report further shows that the long-window epicentre catalogue is partly a record of instrumentation history rather than of seismicity, and separates three physically distinct seismic regimes within the study area.

Keywords: seismic hazard, Modified Mercalli Intensity, ShakeMap, digital elevation model, catalogue completeness, Gutenberg–Richter, terrain visualisation, Pakistan

1. Introduction

Pakistan occupies one of the most tectonically complex regions on Earth. The northward convergence of the Indian plate against Eurasia is accommodated across several structurally distinct systems: continental collision and thrusting along the Himalayan front and the Hazara–Kashmir syntaxis; oblique strike-slip transpression along the Chaman fault system and the Sulaiman–Kirthar fold-and-thrust belt to the west; oceanic subduction beneath the Makran margin in the south; and a deep, persistently active intermediate-depth seismic nest beneath the Hindu Kush to the north-west. These systems differ in depth, mechanism, recurrence and, critically, in the population they place at risk.

Two questions are frequently conflated in popular seismic cartography. The first — where do earthquakes originate? — is answered by an epicentre catalogue. The second — where has the ground actually shaken? — requires a ground-motion field, because intensity decays with distance from the source. A map answering the first question systematically understates exposure in low-seismicity, high-population terrain. This

report documents products addressing both, states their provenance and processing precisely, and analyses what the combination reveals.

2. Data Sources

All inputs are openly licensed and were retrieved programmatically, so the pipeline is reproducible from a clean checkout. Table 1 records provenance, resolution and volume for each input.

Dataset	Provider / product	Resolution & extent	Volume retrieved	Access
Elevation and bathymetry	AWS Terrain Tiles (Mapzen “terrarium” encoding), zoom level 9	~265 m at source; mosaicked to 7168 × 7168 px, resampled to a 4200 × 4122 equal-ground-pixel grid at ~435 m	784 tiles; elevation range −4,399 m (Arabian Sea floor) to +8,181 m	13 Aug 2026
Earthquake catalogue	USGS ANSS Comprehensive Catalog (ComCat) via the FDSN event web service	Point origins: longitude, latitude, depth, magnitude, origin time; frame 59.6–78.4°E, 21.2–37.4°N	14,153 records retrieved; 12,715 of type “earthquake” analysed (1902–2025, M 2.6–8.1)	13 Aug 2026
Ground-motion fields	USGS ShakeMap and ShakeMap Atlas 2.0 (grid.xml products)	~0.033° (~3.3 km) native raster per event; composited by cell-wise maximum	309 event grids, 2000–2025, M ≥ 4.5 (~3 GB of source XML)	13 Aug 2026
National boundary	UN OCHA / Humanitarian Data Exchange, Pakistan ADM0	Vector polygon, EPSG:4326	1 feature (plus ADM1–ADM3 available)	13 Aug 2026
Felt-report validation	USGS “Did You Feel It?” community intensity	Geocoded citizen reports aggregated to Community Decimal Intensity	417 reports for the 2005 Kashmir event; peak CDI 9.1	13 Aug 2026

Table 1. Input datasets, provenance and volume.

Two provenance points deserve emphasis. First, a file in the project named `pakistan_dem_30m.tif` was found not to be 30 m data — its true posting is approximately 1 km — and its southern limit at 23.5°N truncates the coastline, excluding the Arabian Sea. It was therefore replaced. Second, the terrarium tile set carries genuine bathymetry, which is what allows the sea to be rendered as a physical surface rather than a masked constant.

3. Software Stack

The pipeline runs on CPython 3.14 (invoked through the Windows py launcher; the interpreter first on PATH lacks the geospatial stack). Table 2 lists the libraries and their role.

Component	Version	Role in the pipeline
NumPy	2.4.6	Array backbone: DEM mosaicking, hillshade, texture compositing, the 3-D rasteriser
SciPy	1.17.1	Gaussian filtering (relief weighting, feathering) and map_coordinates resampling
Rasterio	1.5.0	GeoTIFF I/O; affine transforms; rasterisation of vector boundaries
GeoPandas / Shapely	1.1.3 / 2.1.2	Boundary geometry handling and buffering for the border line
Pillow	12.2.0	Supersampled downsampling, typography, legend rendering, export
Requests	2.32.5	Concurrent retrieval of terrain tiles, catalogue queries and ShakeMap grids
DuckDB	1.5.5	In-process SQL over the catalogue CSV for the statistical analysis
Matplotlib	3.10.9	Analytical figure (Figure 3)
forge3d	1.35.0	Evaluated for the 3-D pass; not used — see Section 3.1
Hardware	—	Qualcomm Adreno X1-85 (Vulkan backend), 15.6 GB system RAM

Table 2. Software stack.

3.1 Renderer selection

forge3d was the intended 3-D engine and was tested directly rather than assumed. Two independent findings ruled it out on this hardware. Its MapScene recipe does not honour the elevation angle of its OrbitCamera: renders at 20° and 60° differed by a mean of 1.2 levels out of 255, i.e. it produces a flat, top-down draped map. Separately, its path-traced terrain entry point (hybrid_render_terrain_reference) failed on four of four attempts with the error “ReSTIR reuse chain produced no valid reservoirs”, at roughly 60 s per attempt, and required more than 120 s even at 768 × 512. Neither path can produce the oblique, height-displaced geometry the design requires.

A purpose-built perspective terrain rasteriser was therefore written. It uses a painter's algorithm, iterating the elevation grid from the far (northern) edge toward the camera and splatting each row with a vertical skirt that fills inter-row gaps, giving correct occlusion of distant terrain by nearer peaks. It renders a 4K frame in approximately three seconds, against minutes or outright failure for the alternatives.

4. Processing Pipeline

The pipeline is implemented in `Pak_Earthquakes_3D.py`, with two layer modes selected at run time (`--mode epicentres` and `--mode shaking`) sharing a single terrain and composition path. Intermediate products are cached so that look development iterates in seconds rather than re-downloading gigabytes.

4.1 Elevation surface

784 terrarium tiles were fetched concurrently and decoded by the standard relation $\text{elevation} = (R \times 256 + G + B \div 256) - 32768$ metres. The Web Mercator mosaic was then resampled by bilinear interpolation onto a regular longitude/latitude grid whose cell dimensions were chosen so that ground distance per pixel is equal in both axes at the mid-latitude of the frame, avoiding the vertical stretch Mercator would otherwise impose at 37°N.

4.2 Relief shading

Three hillshades (azimuth 315°/270°/15°) were combined, with the vertical exaggeration factor scaled by cell size so that the rendered appearance is invariant to working resolution. The result was contrast-stretched about a bright pivot and then gated by a local-relief weight derived from the residual between the DEM and a 6 km Gaussian, so that alluvial plains remain near-white while mountains take the full tonal range. An ambient-occlusion term derived from the same residual darkens valley floors, giving ranges tonal mass rather than isolated ridge lines.

4.3 Epicentre layer

Each catalogued event was splatted as a radially symmetric Gaussian kernel whose radius and amplitude scale with magnitude; the accumulated field was normalised at its 97.5th percentile and mapped through a five-stop violet ramp. This layer represents epicentre concentration only.

4.4 Shaking layer

For each of the 309 events with a ShakeMap product, the `grid.xml` intensity raster was parsed, reshaped (latitude-descending, longitude-ascending row order) and bilinearly resampled onto the working grid; the running cell-wise maximum was retained. The result is the strongest MMI each location is modelled to have experienced since 2000. Because individual ShakeMap grids are finite rectangles, their union has step discontinuities outside the densely covered core; coverage within Pakistan is complete (zero uncovered cells), so the layer is clipped to national territory with a feathered edge, and this clipping is stated on the map itself.

4.5 Oblique projection and composition

The shaded texture is draped on the elevation grid and projected through a pinhole camera pitched 50° below horizontal, looking due north. Height displacement is

deliberately modest ($\times 13$ on a smoothed height field): the visual drama comes from the hillshade, and heavier extrusion resolves isolated Karakoram summits into needle artefacts. Rendering is done at twice output resolution and Lanczos-downsampled. A directional periwinkle cast shadow is accumulated beneath the landmass, and the plate is finished with serif typography and an explicit legend.

Earthquakes (1900–2025) with
magnitude 2.5 or higher

PAKISTAN

EPICENTRE DENSITY (WHERE RUPTURES BEGAN)

isolated clustered dense
Colour = concentration of epicentres, i.e. the map point above
where each rupture started. It is NOT a measure of shaking:
a place can be violently shaken by a distant earthquake and
still appear white here. See the companion intensity map.

Seismicity: U.S. Geological Survey, Earthquake Hazards Program.
14,153 events of $M \geq 2.5$, 1900–2025.
Elevation & bathymetry: AWS Terrain Tiles (Mapzen/terrarium).
Boundary: UN OCHA / HDX Pakistan ADMs.

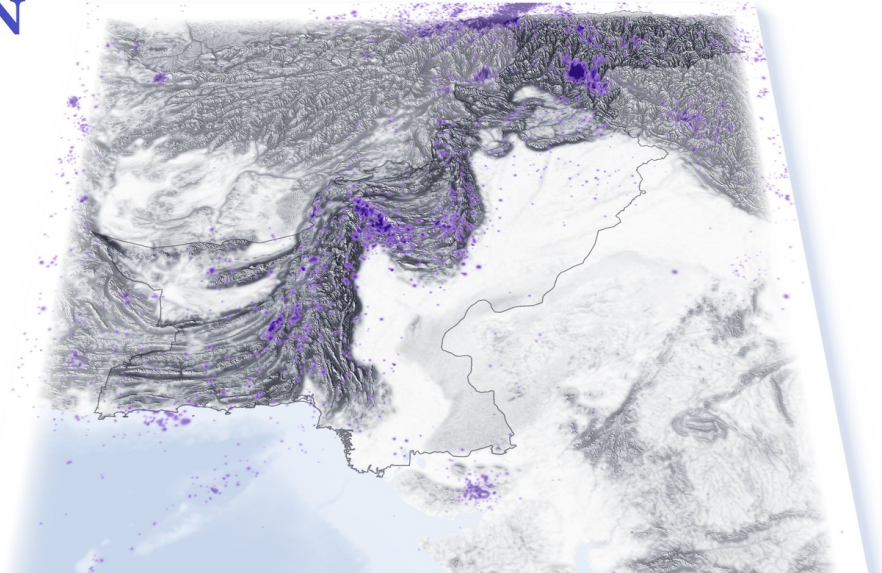


Figure 1. Epicentre density, $M \geq 2.5$, 1902–2025. Colour marks where ruptures began. The Indus plain reads as empty because few earthquakes originate there — not because it is unshaken.

Ground shaking felt (2000–2025):
strongest intensity on record

PAKISTAN

MODIFIED MERCALLI INTENSITY (MAXIMUM EXPERIENCED)

IV V VI VII VIII+
Light · Moderate · Strong · Very strong · Severe
Colour = how hard the ground shook at that place, not where
the earthquake began. Shaking decays with distance from the
source, so the plains are coloured even where no quake starts.

Shaking: USGS ShakeMap / ShakeMap Atlas, cell-wise maximum MMI.
309 events of $M \geq 4.5$, 2000–2025.
Elevation & bathymetry: AWS Terrain Tiles (Mapzen/terrarium).
Boundary: UN OCHA / HDX Pakistan ADMs.

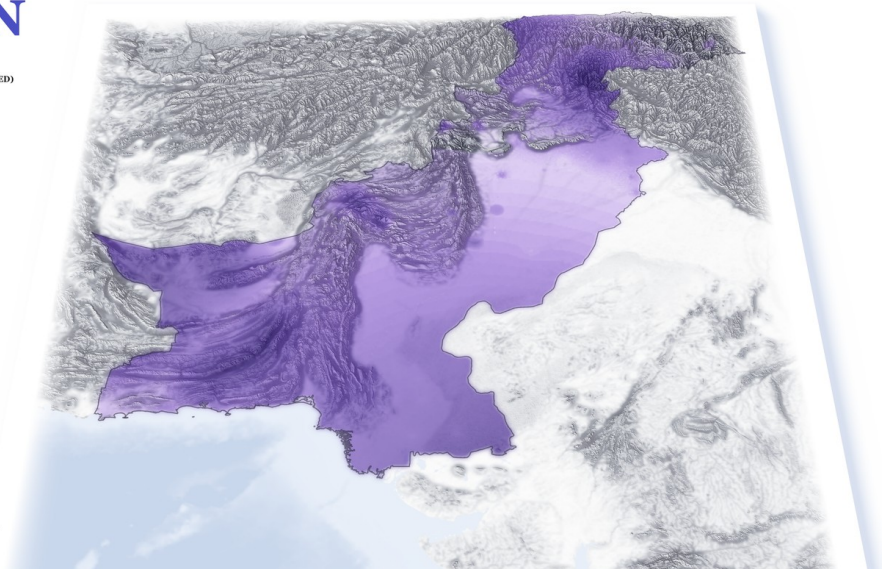


Figure 2. Maximum Modified Mercalli Intensity experienced, 2000–2025, composited from 309 USGS ShakeMap grids. The populated lowlands are now coloured, because intensity decays with distance rather than stopping at the epicentre.

5. Analysis: What the Maps Reveal

5.1 The long catalogue is partly a history of instruments

Before interpreting any pattern, the catalogue must be read as an observational record rather than a physical one. Between 1900 and 1972 the smallest event recorded in the frame in a given year never falls below M 5.2; in 1973, with the maturing of the global digital network, it drops to M 3.1, and annual counts rise by roughly an order of magnitude (Figure 3a). The magnitude–frequency distribution confirms this: the catalogue holds 9,155 events of M 4.0–4.9 but only 2,230 below M 4.0, an inversion of the Gutenberg–Richter expectation [7] that places the completeness magnitude near M 4.5. Consequently the density of the 125-year epicentre map is, in part, a map of detection capability. Restricting analysis to 1990–2025 and $M \geq 4.5$ ($n = 3,154$) gives a maximum-likelihood b-value [8] of 1.22 (Figure 3b) — slightly above the global norm of about 1.0, consistent with a population mixing shallow crustal and deep intra-slab events.

5.2 Three seismic regimes, not one

The depth distribution (Figure 3c) is strongly bimodal and separates the region into physically distinct systems. 41.9 per cent of events are shallower than 35 km — crustal seismicity of the Himalayan front, the Chaman system and the Sulaiman–Kirthar belt. A second population, 27.1 per cent of the catalogue, lies between 150 and 300 km, concentrated in a compact volume beneath the Hindu Kush (3,427 events, mean depth 203 km, maximum M 7.8). This is one of the world's few intermediate-depth seismic nests, sustained by a detaching, sinking slab [11]. Its practical significance is that of the 29 events of $M \geq 7$ in the catalogue, 15 are shallow and 14 are intermediate-depth: roughly half of the largest earthquakes affecting Pakistan originate 150–250 km below the surface, where greater source depth spreads moderate shaking over a very wide footprint instead of concentrating severe shaking locally.

5.3 Origin is not impact — and the gap is where people live

This is the substantive finding, and it is the reason the two maps must be read together. Comparing equal-latitude windows (28–32°N), the Sulaiman–Kirthar belt contains 840 epicentres against 61 in the Punjab plain — a factor of nearly fourteen. On Figure 1 the plain is therefore blank. Yet Figure 2 shows that same plain shaded throughout: 98.4 per cent of Pakistan's land area has experienced at least MMI IV since 2000, 74.6 per cent MMI V or above, 28.9 per cent MMI VI or above, and 7.0 per cent MMI VII or above. The 2005 Kashmir earthquake alone produced a modelled MMI of 5.4 in Lahore, 340 km from its epicentre, and 6.7 in Islamabad.

Table 3 makes the divergence concrete. Every major population centre has a maximum recorded intensity of at least MMI IV–V, and several exceed VI, despite most of them lying far from any epicentre cluster. Karachi's value of 6.3 derives chiefly from distant

sources rather than local ruptures. An epicentre map alone would rank these cities as low-exposure; the intensity field shows otherwise.

City	Max MMI	Descriptor	Principal contributing source
Muzaffarabad	8.6	Severe	2005 Kashmir M 7.6, Balakot–Bagh fault rupture [9]
Islamabad / Rawalpindi	6.7 / 6.3	Strong	2005 Kashmir M 7.6 at ~107 km
Hyderabad	6.5	Strong	Distant Makran and Kachchh sources
Karachi	6.3	Strong	Distant sources, incl. 2001 Bhuj M 7.7
Sukkur	5.9	Moderate	Sulaiman–Kirthar belt events
Peshawar	5.6	Moderate	Hindu Kush intermediate-depth events
Lahore	5.5	Moderate	2005 Kashmir M 7.6 at 340 km
Quetta	5.3	Moderate	Chaman system / 2008 Ziarat sequence
Multan	5.2	Moderate	Sulaiman belt and distant Kashmir sources
Faisalabad	4.8	Light	2005 Kashmir M 7.6 at 350 km

Table 3. Maximum modelled Modified Mercalli Intensity by city, 2000–2025, sampled from the composited ShakeMap field.

5.4 Can intensity be expressed as magnitude?

A natural request is to relabel the intensity field in magnitude units, so that a reader can ask directly “what magnitude did I experience here?”. This cannot be done, and the obstacle is definitional rather than technical. Magnitude is a property of the source: one scalar per earthquake, proportional to the logarithm of the seismic moment released at the rupture. It has no spatial variation — the 2005 Kashmir earthquake was M 7.6 in Muzaffarabad and equally M 7.6 in Karachi. Intensity, by contrast, is a property of a site. Painting a magnitude value onto every grid cell would fabricate a field that does not exist, and would systematically mislead: a cell shaken at MMI V by a distant M 7.6 would be relabelled as though a small local earthquake had occurred there.

The inverse operation is legitimate and is standard practice for pre-instrumental events. Bakun and Wentworth [13] estimate a single source magnitude and location by inverting the entire spatial pattern of reported intensities together with their distances. That procedure consumes the whole field and returns one magnitude per earthquake; it cannot return one per location. Nor does the source data contain a “felt magnitude” variable: ComCat stores a single magnitude per event, and the “Did You Feel It?” system — which is genuinely felt data, contributed by the public — reports Community Decimal Intensity [5], again an intensity, precisely because intensity is the only quantity meaningful at a place.

What is well defined is attribution. During compositing, each cell records not only its maximum intensity but the identity, and therefore the magnitude, of the event that produced it. Figure 4 maps that quantity: the magnitude of the earthquake responsible for the strongest shaking each location has felt since 2000, shown only where at least MMI IV was reached. It answers the intuitive question without violating the definition.

The result is emphatic. Across the 98.4 per cent of national territory that has felt MMI IV or more, 91.3 per cent of that area owes its strongest shaking to an earthquake of M 7.5–8.0, and the median causative magnitude is M 7.7. Lahore's worst episode (MMI 5.5) is attributable to the M 7.6 Kashmir earthquake 340 km away; Multan's and Karachi's to M 7.7 events; Islamabad's and Peshawar's to the same M 7.6 Kashmir rupture. Only a small minority of the country — 4.9 per cent of the felt area, including Quetta, whose maximum of MMI 5.3 came from a local M 6.4 — has its worst experience set by a moderate nearby event rather than a distant great one. Pakistan's felt seismic experience is therefore overwhelmingly a long-range phenomenon.

This also explains why restricting the source catalogue to $M \geq 3$ would not colour the plains: an M 3.0 event is perceptible only within roughly 10–20 km of its epicentre, so a map built from such events would closely resemble Figure 1. The lowlands' felt experience is dominated by large, distant earthquakes, not by small local ones.

The earthquake behind your worst jolt (2000–2025):
magnitude of the causative event

PAKISTAN

MAGNITUDE OF THE EARTHQUAKE THAT SHOOK THIS PLACE HARDEST

M 4.5 M 6.5 M 6.7 M 7.8

Magnitude belongs to the earthquake, not to a place, so intensity cannot be "converted" to magnitude pointwise. What is well defined is attribution: for each location, which event caused its strongest shaking, and how big that event was. Shows only where MMI IV+ was felt.

Shaking: USGS ShakeMap / ShakeMap Atlas, cell-wise maximum MMI.
399 events of M 4.5, 2000–2025.
Elevation & bathymetry: AWS Terrain Tiles (Mapzen/terrarium).
Boundary: UN OCHA / HDX Pakistan ADMo.

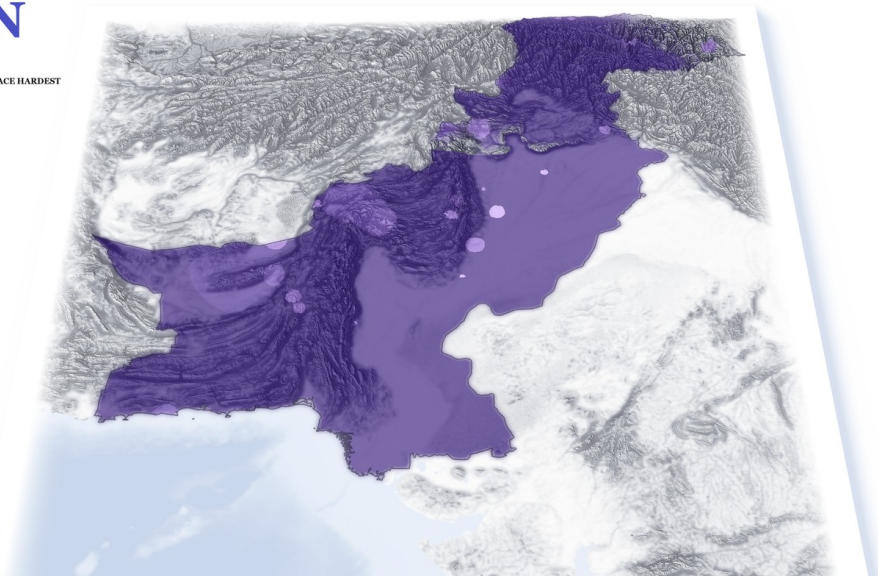


Figure 4. Magnitude of the earthquake responsible for the strongest shaking felt at each location, 2000–2025. Magnitude is attributed from the causative event; it is not a per-location conversion of intensity.

5.5 Implications

Three consequences follow. First, exposure assessment cannot be driven by epicentre proximity: the Indus corridor concentrates population and unreinforced masonry in

exactly the zone that epicentre maps render invisible. Second, because roughly half of the largest regional events are deep Hindu Kush sources whose shaking is broad but moderate, the modal hazard for the plains is repeated MMI IV–VI loading rather than rare catastrophic shaking — a profile that damages non-engineered structures cumulatively. Third, the highest intensities coincide with the mountain belts where relief, slope instability and access constraints compound the direct shaking hazard, as the landslide-dominated losses of 2005 demonstrated. The 25-year window is also short relative to recurrence: the 1945 Makran M 8.1 and the 1935 Quetta M 7.6 both lie outside it, so Figure 2 should be read as a lower bound on credible shaking, not as a maximum-credible-event map.

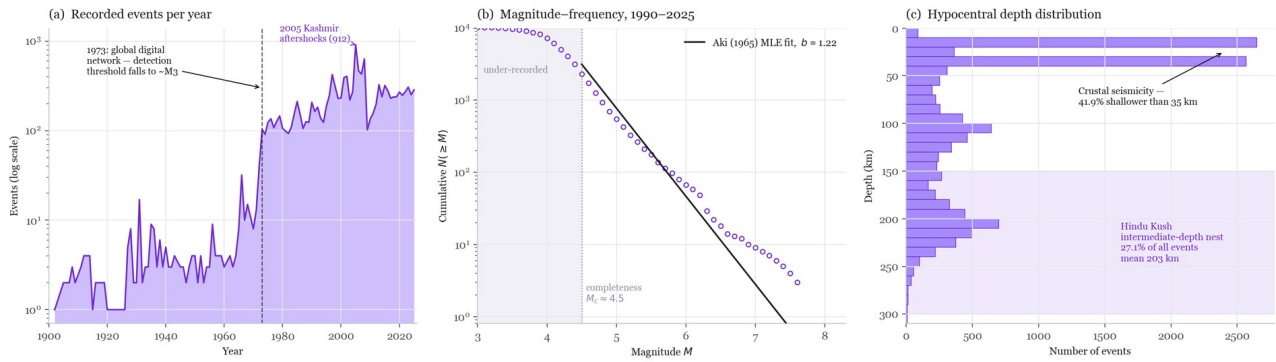


Figure 3. (a) Annual event counts, log scale, showing the 1973 detection discontinuity. (b) Cumulative magnitude-frequency distribution with the Aki maximum-likelihood fit above $M_c \approx 4.5$. (c) Hypocentral depth distribution, isolating the Hindu Kush intermediate-depth nest.

6. Limitations

The intensity field is modelled, not measured. ShakeMap interpolates sparse instrumental and macroseismic observations through ground-motion and intensity prediction equations [2], [4]; in a region with limited strong-motion instrumentation, values far from the source carry substantial uncertainty and do not resolve site-specific soil amplification. The cell-wise maximum discards recurrence entirely: a location shaken once at MMI VI is rendered identically to one shaken twenty times. The composite covers 2000–2025 only, and it is clipped to national territory for coverage reasons, so cross-border continuity is not depicted. The epicentre map's 125-year window is heterogeneous, as Section 5.1 sets out. Finally, the terrain is a visualisation surface rather than an analytical one: vertical exaggeration is $\times 13$ and the height field is smoothed before displacement, so no elevation should be read off the render.

7. Conclusion

Two maps of the same country and the same physical process produce opposite impressions of who is at risk, and the difference is entirely a matter of what is being mapped. Epicentre density describes the tectonics; intensity describes the human exposure. For Pakistan, where the population is concentrated in a lowland corridor flanked by the seismic sources rather than sitting on them, only the second is a

defensible basis for exposure reasoning. Reporting either without an explicit legend distinguishing origin from impact invites precisely the misreading that prompted this work.

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